

MTH 419 Intro to Abstract Algebra

SAMPLE EXAM 1

Name:

SAMPLE

UB Person Number:

0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9

Instructions:

- Each problem is worth 20 points.
- You may use course lecture notes and your own notes.
- No textbooks and electronic devices.
- Use backside pages for scratch work only.
- For full credit solve each problem fully, showing all relevant work.

1. Let $G = \langle a \rangle$ be a cyclic group of order 30, generated by an element a .

a) How many different subgroups does G have? Justify your answer.

b) Let H be a subgroup of G of order 10. Write all possible generators of H in the form a^k . Justify your answer.

2. Consider the following element of the symmetric group S_9 :

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 4 & 1 & 5 & 2 & 7 & 3 & 6 & 9 & 8 \end{bmatrix}$$

Answer the following questions, justifying your answers.

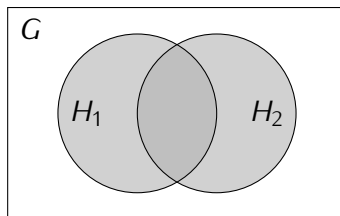
a) What is the order of α ?

b) Is α an even or an odd permutation?

c) Give an example of an element $\beta \in S_9$ such that $|\beta| = 20$. Write β in the matrix notation.

3. Show that there does not exist a group G , such that exactly one element of G has order 3.

4. Let G be a group and let H_1, H_2 be subgroups of G , such that H_1 is not a subset of H_2 and H_2 is not a subset of H_1 . Let $A \subseteq G$ consist of all elements of H_1 and H_2 (i.e. $A = H_1 \cup H_2$). Show that A is not a subgroup of G .



5. Let G, H be groups and let $K \subseteq G \times H$ be a subgroup of their direct product. Define

$$S = \{g \in G \mid (g, h) \in K \text{ for some } h \in H\}$$

In other words, S consists of all elements of G that appear as the first coordinate of some element of K . Show that S is a subgroup of G .